

# Pool equilibria including strategic storage



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## HIGHLIGHTS

- Energy storage systems (ESSs) are considered as price makers in the energy market.
- A multi-period EPEC is proposed to study strategic behaviors of various generators.
- The impacts of different ESSs on the market equilibrium are thoroughly compared.
- Reformulation technique is used to handle the nonlinear complementarity constraints.

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## ABSTRACT

With a rapid increase in capacity, independently-invested energy storage systems (ESSs) might take up a significant share in the generation mix, and would be required to participate in the electricity market. Considering the arbitrage capability of ESSs, their behaviors would be remarkably different from those of conventional generators, impacting the market equilibrium. These impacts would vary with the type of ESSs and the generation mix. Therefore, this paper formulates a multi-period market equilibrium problem with equilibrium constraints (EPEC) to study the strategic behaviors of different types of ESSs and their impacts on the market outcomes, assuming that ESSs behave as price-makers. The EPEC model is established within a general framework, which considers the individual profit-maximization behaviors of different ESSs and generators, including thermal units, hydro units and renewable units. Interactions between different generators and the market operator are also represented. Finally, numerical studies based on a modified IEEE 57-node system with different wind generation curves are performed for illustration and validation.

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## 1. Introduction

At present, energy storage systems (ESSs) play an increasingly important role in supporting the secure and economic operation of power systems, especially in case of large-scale renewable penetration [1,2]. In the foreseeable future, the share of renewables and ESSs, which might belong to different companies, may rapidly increase, making producers based on these technologies major market players. Thus, privately-owned and independently-invested wind farms, solar plants and ESSs would be required to participate in the electricity market just as conventional generators do. In this case, all generators will jointly compete in the market and strategically pursue their respective maximum profits with potential market power. Considering the arbitrage capability of

ESSs, their behaviors would be remarkably different from those of conventional generators, impacting differently the market equilibrium. And these impacts would vary with the type of ESSs and the generation mix.

Several papers have studied the participation of ESSs in electricity markets. However, most of these papers simply regard ESSs as price-takers [3], or as a part of a virtual power plant, but not as individual price-makers. Ref. [4] proposes a mixed complementarity model to study the impacts of ESS sizes and locations on a perfectly competitive market. With a forecasted, fixed hourly marginal clearing price (MCP) curve, the bidding strategies for pumped hydroelectric storage (PHS) are analyzed in [5]. Strategies for the integrated self-scheduling of a wind farm and a PHS are proposed in [6]. Robust bidding in combination with ESS is presented in [7], which uses forecast errors of MCPs and wind generation. An optimal bidding mechanism for an independent ESS is proposed in [8], assuming the ESS behavior to be price-taker. Moreover, revenue analysis for ESSs participating in the electricity markets are

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studied in [9–12], but without considering the impacts of ESSs on MCPs.

In fact, in order to perform an equilibrium analysis of a pool-based electricity market with ESSs, it is necessary to extend the market equilibrium model from single-period to multi-period, representing that the ESS intra-day operation state may switch from charging to discharging and vice versa. Available models pay limited attention to the multi-period market equilibrium analysis and to ESS strategic arbitrage behavior. Clearly, extending the analysis to a multi-period framework significantly increases the complexity of the market equilibrium problem, which is an equilibrium problem with equilibrium constraints (EPEC). Most of the related papers concentrate on a single-period EPEC with thermal units [13–20], or use a Nash-Cournot equilibrium model rather than a supply function equilibrium model [21–23], or find a near-equilibrium under locational marginal pricing and minimum profit conditions [24]. Moreover, the differences in technical characteristics of ESSs have significant influences on their bidding strategies, as different types of ESSs with different energy limits, efficiencies and operation costs have different impacts on the market equilibrium. This issue is not carefully considered in the available literature.

In brief, the research on the impacts of ESSs on the electricity market equilibrium and the investigation of the strategic behaviors of various types of ESSs are limited and need further attention. Therefore, this paper focuses on electricity market equilibria considering the strategic behaviors of various types of ESSs, which is also the main contribution of this paper. First, the ESSs are considered as *price makers* strategically choosing their bidding prices and their discharging or charging strategies. Second, a *multi-period EPEC* is established to study the market equilibrium and the strategic interactions among different generators, including ESSs, thermal units, conventional hydro units, wind farms and solar plants. Third, the impacts of *different types* of ESSs, PHS, compressed air energy storage (CAES) and zinc bromine flow battery storage (ZBFBS) are analyzed to accurately compare their different technical characteristics [2]. The changes in market prices, consumer payments and arbitrage profits are comprehensively studied in a modified IEEE 57-node system.

Several cases are compared to analyze the strategic behaviors of different ESSs. A case with only strategic thermal units, a hydro unit, a wind unit and a solar unit is first considered as a benchmark. Then, the strategic cases respectively involving PHS, CAES or ZBFBS are investigated. For comparison, the non-strategic cases where producers offer at true marginal costs are also studied. In order to focus on the strategic behaviors of various generators, the uncertainties of renewable generation are not included in the EPEC model but different wind generation curves are considered in the case studies.

The rest of this paper is organized as follows. The procedures to formulate the EPEC model are described in Section 2. The equilibrium model of the electricity market is described in Section 3. The MPEC of each agent is derived in Section 4 and the joint Karush–Kuhn–Tucker (KKT) conditions are collected together to form an EPEC in Section 5. Numerical examples to illustrate the impacts of the ESSs on the market equilibrium are provided in Section 6. Finally, conclusions are summarized in Section 7.

## 2. EPEC modeling

In a bid-based electricity market, all participants are leaders and the market operator (MO) that clears the market after receiving all bids is the single-follower, which constitutes a Stackelberg game [18]. Therefore, to formulate the equilibrium problem, a bi-level model per participant is first formulated. This hierarchical

formulation includes each generator's individual profit-maximization problem as part of the upper-level model and the market clearing conditions, incorporating the operation constraints of various generators and the power system, as the lower-level problem. The detailed schematic diagram for the bi-level equilibrium model is shown in Fig. 1(a).

To solve such problem, two general methodologies are often used [19]. The first one is the diagonalization methods that iteratively solve each participant's individual profit-maximization problem until a Nash stationary point is obtained. The second one collects the KKT conditions of all the participants' individual profit-maximization problems and solves them together. Considering the lack of versatility of the diagonalization methods [19–26], the KKT method is adopted in this paper.

Specifically, a mathematical program with equilibrium constraints (MPEC) is derived for each generator by considering the KKT conditions of the lower-level model. Then, to formulate the EPEC, the first-order optimal conditions of each MPEC are derived, and then, all these KKT conditions are jointly considered. The principle of this procedure is presented in Fig. 1(b) [20].

## 3. Model of market equilibrium

We assume that the participants strategically submit their bidding prices to the MO before the gate closure. Then, the MO clears the energy market considering the operational constraints of various generators and the power system to obtain the MCPs and the production levels for each generator.

### 3.1. Upper-level model

The upper-level models are the individual profit-maximization problems of the different generators. The problem of the ESS  $Si$  is, ( $\forall Si = 1, 2, \dots, ES$ ):

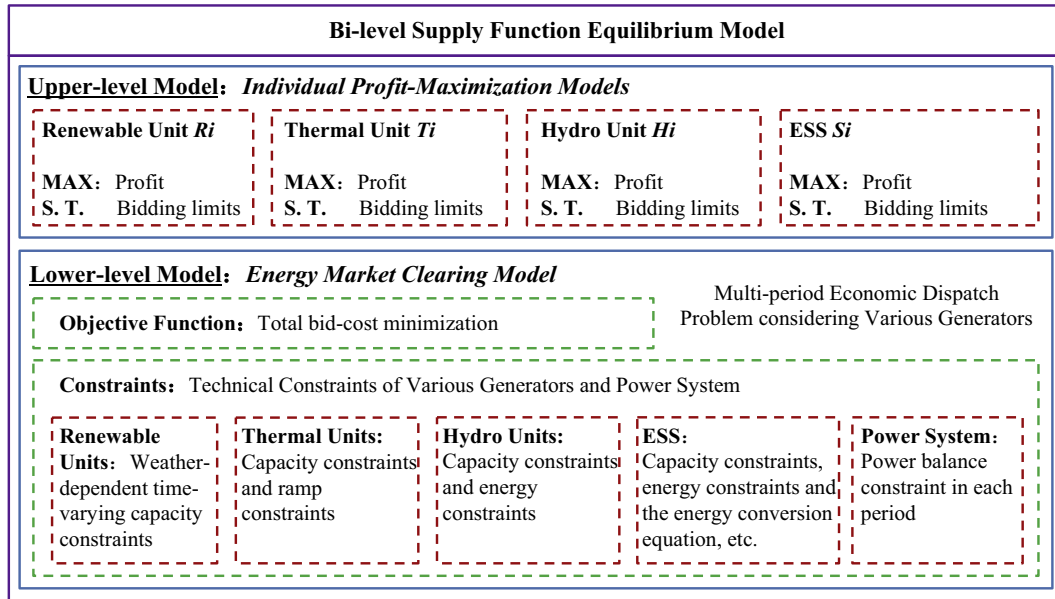
$$\begin{aligned} \min_{\beta_{Si}} \pi_{Si} = & \sum_{t=1}^T \{ b_{Si} [q_{Si}^{cha}(t) + q_{Si}^{dis}(t)] - \lambda(t) [q_{Si}^{dis}(t) - q_{Si}^{cha}(t)] \} \\ & \begin{cases} \beta_{Si}(t) - \beta_{\min} \geq 0 \\ \beta_{\max} - \beta_{Si}(t) \geq 0 \end{cases} \end{aligned} \quad (1-a)$$

where  $b_{Si}$  is the variable non-fuel operation and maintenance (O&M) costs of  $Si$ ,  $q_{Si}^{cha}(t)$  and  $q_{Si}^{dis}(t)$  are the charging and discharging power which are determined by the MO in period  $t$ ,  $\lambda(t)$  is the uniform MCP in period  $t$ ,  $\beta_{Si}(t)$  is the bidding price of  $Si$  strategically submitted to MO,  $\beta_{\min}$  and  $\beta_{\max}$  are the floor and cap bidding prices, respectively, which are determined by MO. The problem of capital costs or fixed costs recovery is related to the capacity market or pricing mechanism regarding capacity, which is beyond the scope of the paper.

The individual profit-maximization models of a thermal unit ( $\forall Ti = 1, 2, \dots, TP$ ), hydro unit ( $\forall Hi = 1, 2, \dots, HP$ ) and renewable unit ( $\forall Ri = 1, 2, \dots, RE$ ) can be formulated as:

$$\begin{aligned} \min_{\beta_{Ti}} \pi_{Ti} = & \sum_{t=1}^T [a_{Ti} q_{Ti}^2(t) + b_{Ti} q_{Ti}(t) - \lambda(t) q_{Ti}(t)] \\ & \begin{cases} \beta_{Ti}(t) - \beta_{\min} \geq 0 \\ \beta_{\max} - \beta_{Ti}(t) \geq 0 \end{cases} \end{aligned} \quad (1-b)$$

$$\begin{aligned} \min_{\beta_{Hi}} \pi_{Hi} = & \sum_{t=1}^T [b_{Hi} - \lambda(t)] q_{Hi}(t) \\ & \begin{cases} \beta_{Hi}(t) - \beta_{\min} \geq 0 \\ \beta_{\max} - \beta_{Hi}(t) \geq 0 \end{cases} \end{aligned} \quad (1-c)$$



(a) Bi-level Equilibrium Model

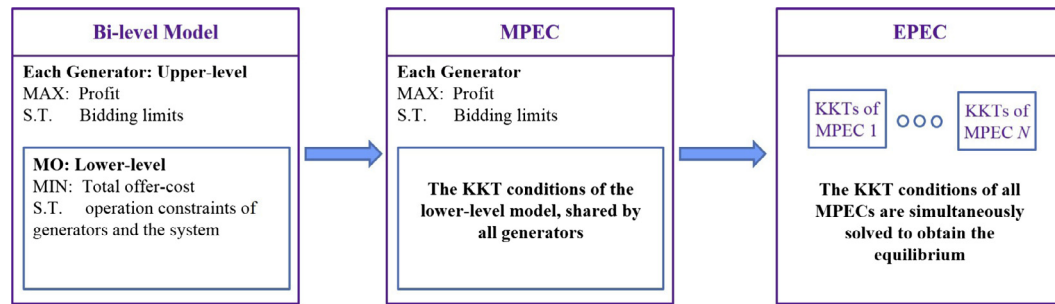


Fig. 1. Flow-chart for the supply function equilibrium problem.

$$\min_{\beta_{Ri}} \pi_{Ri} = \sum_{t=1}^T [b_{Ri} - \lambda(t)] q_{Ri}(t) \quad (1-d)$$

$$\begin{cases} \beta_{Ri}(t) - \beta_{\min} \geq 0 \\ \beta_{\max} - \beta_{Ri}(t) \geq 0 \end{cases}$$

where  $a_{Ti}, b_{Ti} \neq 0$  are variable generation cost coefficients of the thermal unit  $T_i$ ,  $b_{Hi}, b_{Ri}$  are the variable O&M costs of hydro unit  $H_i$  and renewable unit  $R_i$ , while  $q_{Ti}(t), q_{Hi}(t), q_{Ri}(t)$  are the assigned generation amounts to  $T_i, H_i$  and  $R_i$ , respectively,  $\beta_{Ti}(t), \beta_{Hi}(t)$  and  $\beta_{Ri}(t)$  are bidding prices of  $T_i, H_i$  and  $R_i$ , in period  $t$ .

### 3.2. Lower-level model

The market-clearing lower-level model is described below. The objective function is to minimize the total bid-costs:

$$\min F = \sum_{t=1}^T \sum_{i=1}^{TP} [a_{Ti} q_{Ti}(t) + \beta_{Ti}(t)] q_{Ti}(t) + \sum_{t=1}^T \sum_{i=1}^{HP} \beta_{Hi}(t) q_{Hi}(t) + \sum_{t=1}^T \sum_{i=1}^{RE} \beta_{Ri}(t) q_{Ri}(t) + \sum_{t=1}^T \sum_{i=1}^{ES} \beta_{Si}(t) [q_{Si}^{dis}(t) + q_{Si}^{cha}(t)] \quad (2-a)$$

where  $\beta_{Ti}(t), \beta_{Hi}(t), \beta_{Ri}(t), \beta_{Si}(t)$  are the intercepts of the supply curves (that is, bidding prices) strategically submitted by generators to maximize their own profits [27]. The slope of the supply curve of thermal unit  $T_i$  is assumed to be  $a_{Ti}$  while those of hydro units,

renewable units and the ESSs are considered zero.  $TP, HP, RE$  and  $ES$  are the total numbers of thermal units, hydro units, renewable units and ESSs.

Moreover, the technical operation constraints of various generators should be included in the lower-level model and considered by the MO because the production levels of participants are determined by the MO considering their strategic bidding prices.

#### 3.2.1. Technical constraints of the ESSs

$\forall t \in T, \forall Si \in ES$ , the discharging and charging power limits, energy constraints and the energy conversion equation are [4]:

$$\begin{cases} q_{Si}^{dis}(t) \geq 0 : \mu_{Si}^{dis, \min}(t) \geq 0 \\ q_{Si \max} - q_{Si}^{dis}(t) \geq 0 : \mu_{Si}^{dis, \max}(t) \geq 0 \\ q_{Si}^{cha}(t) \geq 0 : \mu_{Si}^{cha, \min}(t) \geq 0 \\ q_{Si \max} - q_{Si}^{cha}(t) \geq 0 : \mu_{Si}^{cha, \max}(t) \geq 0 \\ E_{Si}(t) \geq 0 : \mu_{Si}^{E, \min}(t) \geq 0 \\ E_{Si \max} - E_{Si}(t) \geq 0 : \mu_{Si}^{E, \max}(t) \geq 0 \\ E_{Si}(t+1) - E_{Si}(t) + q_{Si}^{dis}(t)/\eta_{Si}^{dis} - \eta_{Si}^{cha} q_{Si}^{cha}(t) = 0 : \lambda_{Si}^{con}(t) \\ E_{Si|end} - E_{Si|init} = 0 : \lambda_{Si}^{bal} \end{cases} \quad (2-b)$$

where  $q_{Si \max}$  is the maximum capacity of the ESS  $S_i$ ,  $E_{Si}(t)$  is the stored energy of ESS  $S_i$  in period  $t$ ,  $E_{Si \max}$  is the maximum energy

that the ESS  $Si$  can store,  $\eta_{Si}^{dis}$  and  $\eta_{Si}^{cha}$  are the discharging and charging efficiencies of the ESS  $Si$ . Note that the last equation of (2-b) ensures that the final state of charge equals to the initial state of charge on a daily cycle.

### 3.2.2. Technical constraints of the thermal units

$\forall t \in T, \forall Ti \in TP$ , the capacity constraints and the ramp constraints are:

$$\begin{cases} q_{Ti}(t) - q_{Ti\min} \geq 0: & \mu_{Ti}^{\min}(t) \geq 0 \\ q_{Ti\max} - q_{Ti}(t) \geq 0: & \mu_{Ti}^{\max}(t) \geq 0 \\ q_{Ti}^{RMPU} - q_{Ti}(t+1) + q_{Ti}(t) \geq 0: & \mu_{Ti}^{RMPU}(t) \geq 0 \\ q_{Ti}^{RMPD} - q_{Ti}(t) + q_{Ti}(t+1) \geq 0: & \mu_{Ti}^{RMPD}(t) \geq 0 \end{cases} \quad (2-c)$$

where  $q_{Ti\min}$  and  $q_{Ti\max}$  are the minimum and maximum capacity limits of the thermal unit  $Ti$ ,  $q_{Ti}^{RMPU}$  and  $q_{Ti}^{RMPD}$  are the maximum ramp up and ramp down limits the thermal unit  $Ti$ .

### 3.2.3. Technical constraints of the hydro units

$\forall t \in T, \forall Hi \in HP$ , the capacity constraints and the energy constraint considering the water resource limit are:

$$\begin{cases} q_{Hi}(t) - q_{Hi\min} \geq 0: & \mu_{Hi}^{\min}(t) \geq 0 \\ E_{Hi\max} - \sum_{t=1}^T q_{Hi}(t) \geq 0: & \mu_{Hi}^E \geq 0 \\ q_{Hi\max} - q_{Hi}(t) \geq 0: & \mu_{Hi}^{\max}(t) \geq 0 \end{cases} \quad (2-d)$$

where  $q_{Hi\min}$  and  $q_{Hi\max}$  are the minimum and maximum capacity limits of the hydro unit  $Hi$ ,  $E_{Hi\max}$  is the maximum energy that the hydro unit  $Hi$  can provide during the whole day.

### 3.2.4. Technical constraints of the renewable units

$\forall t \in T, \forall Ri \in RE$ , the weather-dependent time-varying capacity constraints are:

$$\begin{cases} q_{Ri}(t) - q_{Ri\min}(t) \geq 0: & \mu_{Ri}^{\min}(t) \geq 0 \\ q_{Ri\max}(t) - q_{Ri}(t) \geq 0: & \mu_{Ri}^{\max}(t) \geq 0 \end{cases} \quad (2-e)$$

where  $q_{Ri\min}(t)$  and  $q_{Ri\max}(t)$  are the minimum and maximum generation limits based on the forecasted wind speed or solar irradiation. Note that the minimum generation limit is usually zero and the actual renewable generation may not reach the maximum limit because of the high variabilities of renewables and limited flexibilities of the power system.

### 3.2.5. Power balance

$\forall t \in T$ , the power balance constraint in period  $t$  is:

$$\begin{aligned} & \sum_{i=1}^{TP} q_{Ti}(t) + \sum_{i=1}^{HP} q_{Hi}(t) + \sum_{i=1}^{RE} q_{Ri}(t) + \sum_{i=1}^{ES} q_{Si}^{dis}(t) - \sum_{i=1}^{ES} q_{Si}^{cha}(t) \\ & - q_D(t) \\ & = 0: \quad \lambda(t) \end{aligned} \quad (2-f)$$

where  $q_D(t)$  is the electricity demand in period  $t$ . Transmission limits and the uncertainty of renewables are not included in the model for simplicity, but different wind generation curves will be considered in case studies. The primal variables of the lower-level model are  $q_{Si}^{dis}(t)$ ,  $q_{Si}^{cha}(t)$ ,  $q_{Ti}(t)$ ,  $q_{Hi}(t)$  and  $q_{Ri}(t)$ , while the dual multipliers of the corresponding constraints are  $\mu_{Si}^{dis,\min}(t)$ ,  $\mu_{Si}^{dis,\max}(t)$ ,  $\mu_{Si}^{cha,\min}(t)$ ,  $\mu_{Si}^{cha,\max}(t)$ ,  $\mu_{Si}^{E\max}(t)$ ,  $\mu_{Si}^{E\min}(t)$ ,  $\mu_{Si}^{con}(t)$ ,  $\lambda_{Si}^{bal}$ ,  $\mu_{Ti}^{\min}(t)$ ,  $\mu_{Ti}^{\max}(t)$ ,  $\mu_{Ti}^{RMPU}(t)$ ,  $\mu_{Ti}^{RMPD}(t)$ ,  $\mu_{Hi}^{\min}(t)$ ,  $\mu_{Hi}^E$ ,  $\mu_{Hi}^{\max}(t)$ ,  $\mu_{Ri}^{\min}(t)$ ,  $\mu_{Ri}^{\max}(t)$  and  $\lambda(t)$ .

## 4. Individual single-level optimization model-MPEC

To transform the bi-level model of each market participant into a single-level model (MPEC), the lower-level problem is replaced by its KKT conditions. Considering the limited space, a generic mathematical formulation is adopted in this section.

### 4.1. Generic mathematical formulation

The generic formulation for the upper-level problem of ESS  $Si$ , thermal unit  $Ti$ , hydro unit  $Hi$  and renewable unit  $Ri$  is:

$$\begin{aligned} & \min \pi_{Xi}[\beta_{Xi}(t)] \\ & \text{s.t. } \mathbf{G}_{Xi}[\beta_{Xi}(t)] \geq \mathbf{0} \end{aligned} \quad (3-a)$$

where  $\pi_{Xi}$  is the objective function and  $\mathbf{G}_{Xi}$  are the sets of bidding price limits, and  $X \in \{S, T, H, R\}$ .

The generic formulation for the lower-level problem is:

$$\begin{aligned} & \min F[q_{Xi}(t)] \\ & \text{s.t. } \begin{cases} \mathbf{H}[q_{Xi}(t)] \geq \mathbf{0}: & \mu \geq \mathbf{0} \\ \mathbf{Y}[q_{Xi}(t)] = \mathbf{0}: & \lambda \end{cases} \end{aligned} \quad (3-b)$$

where  $F$  is the objective function of total bid-cost minimization,  $\mathbf{H}$  and  $\mathbf{Y}$  are the sets of operational constraints of different generators and the power balance constraint,  $q_{Xi}(t) \in \mathbf{q}_X(t) = \{q_{Si}^{dis}(t), q_{Si}^{cha}(t), q_{Ti}(t), q_{Hi}(t), q_{Ri}(t)\}$ .

The Lagrange function  $L$  of the lower-level problem is:

$$L = F[q_{Xi}(t)] - \mu \cdot \mathbf{H}[q_{Xi}(t)] - \lambda \cdot \mathbf{Y}[q_{Xi}(t)] \quad (3-c)$$

and the KKT conditions of the lower-level problem are:

$$\begin{cases} \nabla_{q_{Xi}(t)} F[q_{Xi}(t)] - \mu \cdot \nabla_{q_{Xi}(t)} \mathbf{H}[q_{Xi}(t)] - \lambda \cdot \nabla_{q_{Xi}(t)} \mathbf{Y}[q_{Xi}(t)] = \mathbf{0} \\ \mathbf{Y}[q_{Xi}(t)] = \mathbf{0} \\ \mathbf{0} \leq \mu \perp \mathbf{H}[q_{Xi}(t)] \geq \mathbf{0} \end{cases} \quad (3-d)$$

### 4.2. Reformulation of the complementarity constraints

Among the KKT conditions, the complementarity conditions are highly nonlinear and generally ill-conditioned. In order to make these complementarity constraints “less nonlinear” and easier to handle, auxiliary slack variables  $\mathbf{s}$  can be introduced in the following way [25]:

$$\mathbf{0} \leq \mu \perp \mathbf{H}[q_{Xi}(t)] \geq \mathbf{0} \iff \begin{cases} \mathbf{H}[q_{Xi}(t)] - \mathbf{s} = \mathbf{0} \\ \mathbf{0} \leq \mu \perp \mathbf{s} \geq \mathbf{0} \end{cases} \quad (3-e)$$

Then, the Fisher-Burmeister (FB) function is applied to transform the complementarity constraint into a nonlinear equality constraint [28]:

$$\mathbf{0} \leq \mu \perp \mathbf{s} \geq \mathbf{0} \iff FB(\mu, \mathbf{s}) = \sqrt{\mu^2 + \mathbf{s}^2} - (\mu + \mathbf{s}) = \mathbf{0} \quad (3-f)$$

After the above two steps of reformulation, the MPEC is then derived for each market agent.

### 4.3. MPEC

By substituting the KKT conditions of the lower-level model into the upper-level model, the MPEC of each participant is ( $X \in \{S, T, H, R\}$ ):

$$\begin{aligned} & \min \pi_{Xi}[\beta_{Xi}(t), \mathbf{q}_X(t), \mu, \lambda, \mathbf{s}] \\ & \begin{cases} \mathbf{G}_{Xi}[\beta_{Xi}(t)] \geq \mathbf{0}: & \omega_{Xi} \geq \mathbf{0} \\ \mathbf{K} = \nabla_{q_{Xi}(t)} F[q_{Xi}(t)] - \mu \cdot \nabla_{q_{Xi}(t)} \mathbf{H}[q_{Xi}(t)] - \lambda \cdot \nabla_{q_{Xi}(t)} \mathbf{Y}[q_{Xi}(t)] = \mathbf{0}: & \gamma_{Xi} \\ \mathbf{Y}[q_{Xi}(t)] = \mathbf{0}: & \theta_{Xi} \\ \mathbf{H}[q_{Xi}(t)] - \mathbf{s} = \mathbf{0}: & \eta_{Xi} \\ \sqrt{\mu^2 + \mathbf{s}^2} - (\mu + \mathbf{s}) = \mathbf{0}: & \varepsilon_{Xi} \end{cases} \end{aligned} \quad (3-g)$$

## 5. EPEC

The joint consideration of the KKT conditions of the MPEC of each ESS, thermal unit, hydro unit and renewable unit constitutes an EPEC. The Lagrange function  $L_{Xi}$  of  $X_i$ 's MPEC is ( $X \in \{S, T, H, R\}$ ):

$$L_{Xi} = \pi_{Xi} - \omega_{Xi} \cdot \mathbf{G}_{Xi} - \gamma_{Xi} \cdot \mathbf{K} - \theta_{Xi} \cdot \mathbf{Y}[q_{Xi}(t)] - \eta_{Xi} \cdot \{\mathbf{H}[q_{Xi}(t)] - \mathbf{s}\} - \varepsilon_{Xi} \cdot \left[ \sqrt{\boldsymbol{\mu}^2 + \mathbf{s}^2} - (\boldsymbol{\mu} + \mathbf{s}) \right] \quad (4-a)$$

The KKT conditions of the  $X_i$ 's MPEC are ( $X \in \{S, T, H, R\}$ ) presented in (4-b).

$$\begin{cases} \nabla_{\beta_{Xi}(t)} L_{Xi} = 0 \\ \nabla_{\mathbf{q}_X(t)} L_{Xi} = \mathbf{0} \\ \nabla_{\boldsymbol{\mu}} L_{Xi} = \mathbf{0} \\ \nabla_{\lambda} L_{Xi} = \mathbf{0} \\ \nabla_{\mathbf{s}} L_{Xi} = \mathbf{0} \\ \mathbf{K} = \nabla_{q_{Xi}(t)} F[q_{Xi}(t)] - \boldsymbol{\mu} \cdot \nabla_{q_{Xi}(t)} \mathbf{H}[q_{Xi}(t)] - \lambda \cdot \nabla_{q_{Xi}(t)} \mathbf{Y}[q_{Xi}(t)] = \mathbf{0} \\ \mathbf{Y}[q_{Xi}(t)] = \mathbf{0} \\ \mathbf{H}[q_{Xi}(t)] - \mathbf{s} = \mathbf{0} \\ \sqrt{\boldsymbol{\mu}^2 + \mathbf{s}^2} - (\boldsymbol{\mu} + \mathbf{s}) = \mathbf{0} \\ \mathbf{0} \leq \omega_{Xi} \perp \mathbf{G}_{Xi}[\beta_{Xi}(t)] \geq \mathbf{0} \end{cases} \quad (4-b)$$

The EPEC can then be obtained by concatenating all the KKT conditions of each MPEC and then the optimal conditions are solved together to get the market equilibria. It is important to note that the EPEC is nonconvex [29,30] and thus it may have multiple solutions. If there exist multiple equilibria, the equilibrium that can achieve the total generation cost minimization or the total profit maximization of all participants will be selected as the meaningful one [20]. In this paper, we solve the EPEC (4-b) directly using KNITRO 9.0 under GAMS, which can effectively handle the complementarity constraints (last row of (4-b)) [31], and no additional auxiliary optimization problem is needed.

## 6. Numerical examples

### 6.1. Basic data

The numerical analysis is carried out using a modified IEEE 57-node system with a total installed capacity excluding the ESS of 850 MW. The generation mix includes 8 thermal generators (730 MW), 1 conventional hydro plant (20 MW), 1 wind farm (50 MW) and 1 solar plant (50 MW). A 25 MW ESS is included to study its strategic behavior and its impacts on the market equilibrium. The maximum and minimum supply-demand ratios are 184% and 135%, respectively.

The data for the thermal generators are obtained from [32]. The minimum power output and energy limit of the hydro plant are 5 MW and 240 MW h, respectively. The profiles of wind power generation, solar power generation and the load are shown in Fig. 2. The peak load is 650 MW while the average load is 551 MW. The load difference between peak and valley is relatively small and equal to 188 MW. The O&M costs of the hydro, wind and solar units are assumed to be zero [33]. The parameters of the different types of ESSs are given in Table 1 [2,34].

The PHS, underground CAES and ZBFBS are particularly suited for energy management, load shifting and renewables integration. The PHS is a mature storage technology and makes up nearly 99% of the total installed storage capacity in the world [2]. The energy and capacity limits of PHS are normally larger than those of other storage systems. However, the low energy densities and power densities of PHS and the location problem entail a high capital cost

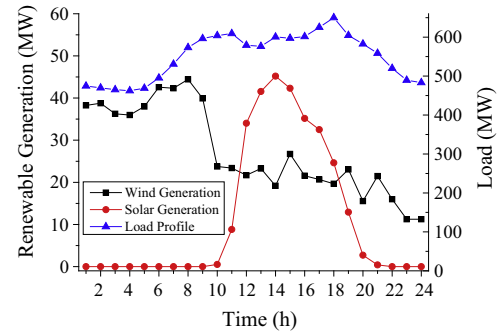


Fig. 2. Load, wind generation and solar generation data for the strategic BC.

Table 1  
Parameters of the ESSs.

| Type  | O&M cost (\$/MW h) | Capacity (MW) | Energy limit (MW h) | Duration (h) | Overall efficiency (%) |
|-------|--------------------|---------------|---------------------|--------------|------------------------|
| PHS   | 0.2                | 25            | 250                 | 10           | 76.5                   |
| CAES  | 2.5                | 25            | 500                 | 20           | 81.0                   |
| ZBFBS | 1.0                | 25            | 125                 | 5            | 60.0                   |

but a low variable cost. The ZBFBS has smaller energy and capacity limits with high energy density and low power density. The total cost of ZBFBS is lower than that of PHS but the O&M cost is higher. Underground CAES has longer duration with lower fixed cost and higher O&M cost as compared to PHS and ZBFBS.

It should be noted that the energy density (kW h/kg or kW h/l) is the value of rated stored energy vs. physical mass or space and the power density (kW/kg or kW/l) is the amount of rated power vs. physical mass or space [35].

The ranking on overall efficiency from top to bottom is CAES, PHS and ZBFBS. It should be noted that the parameters presented in Table 1 are just for the sake of comparison, as the capacity is equivalently chosen as 25 MW for each ESS. The energy limit of any ESS is determined by the product of capacity and duration. The charging or discharging efficiencies of PHS, CAES and ZBFBS are 87.5%, 90.0% and 77.5%, respectively.

Next, a case study with strategic conventional generators and renewable units is considered as a benchmark. Based on this case, the strategic and non-strategic (competitive) cases including PHS, CAES or ZBFBS are comprehensively compared to show the impacts of the arbitrage behaviors of different ESSs on the market equilibrium. Moreover, the strategic case means that all participants are price-makers and submit their bidding prices according to the profit-maximization principle considering the possible reactions of rivals. The non-strategic case means that the energy market is perfectly competitive and all participants submit their true variable generation costs and each generator does not have the ability to change the MCP.

### 6.2. Benchmark case

The strategic benchmark case (BC) is studied without ESSs and the market equilibrium results are illustrated in Figs. 2 and 3. The marginal generator in each period is a thermal plant and hence it determines the MCP. From period 00:00 to 07:00 a.m., the low MCPs are caused by large wind generation and the relatively low demand in the night. From 12:00 to 18:00 p.m., the large solar generation and the hydroelectricity collectively achieve a peak shaving effect on MCP. The expensive thermal plants are replaced by the cost-effective renewable generators and thus the MCPs are comparatively lower in these periods. In periods 10:00 and 11:00 a.m., 19:00 and 20:00 p.m., the MCP is much higher due to the low

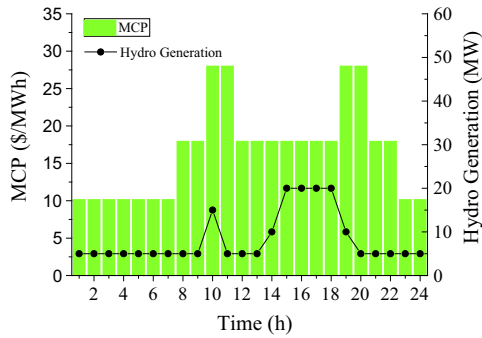


Fig. 3. The MCP and hydro generation results for the strategic BC.

renewable and hydro generation and relatively high demand. Moreover, the MCP curve has significantly positive correlation with the load curve, as shown in Figs. 2 and 3.

### 6.3. Strategic behaviors of various types of ESSs

Considering the 25-MW ESSs of types PHS, CAES and ZBFBS, outcomes show that the main factors that influence the strategic

Table 2

Comparisons of charged and discharged energy of ESSs.

| Cases         |                          | BC<br>+ PHS | BC<br>+ CAES | BC<br>+ ZBFBS |
|---------------|--------------------------|-------------|--------------|---------------|
| Strategic     | Charged energy (MW h)    | 205.7       | 270.9        | 167.4         |
|               | Discharged energy (MW h) | 159.1       | 219.4        | 100.5         |
| Non-strategic | Charged energy (MW h)    | 165.5       | 119.9        | 161.3         |
|               | Discharged energy (MW h) | 126.7       | 97.09        | 96.88         |

behaviors of ESSs are the overall efficiency and the energy limit. In the strategic case, all types of ESSs produce electricity when the MCP is high (08:00–10:00 a.m. and 14:00–18:00 p.m.) and recharge while the MCP is relatively low (00:00–06:00 a.m. and 22:00–24:00 p.m.), as shown in Fig. 4. Specifically, the PHS as well as the hydro unit strategically sell more energy during the 14th periods rather than the 12th and 13th hours because the electricity demands are higher and the total renewable generations are relatively lower.

With higher energy limit and being more efficient, the underground CAES both charges and discharges more than the other two ESSs to strategically arbitrage during the whole day. ZBFBS,

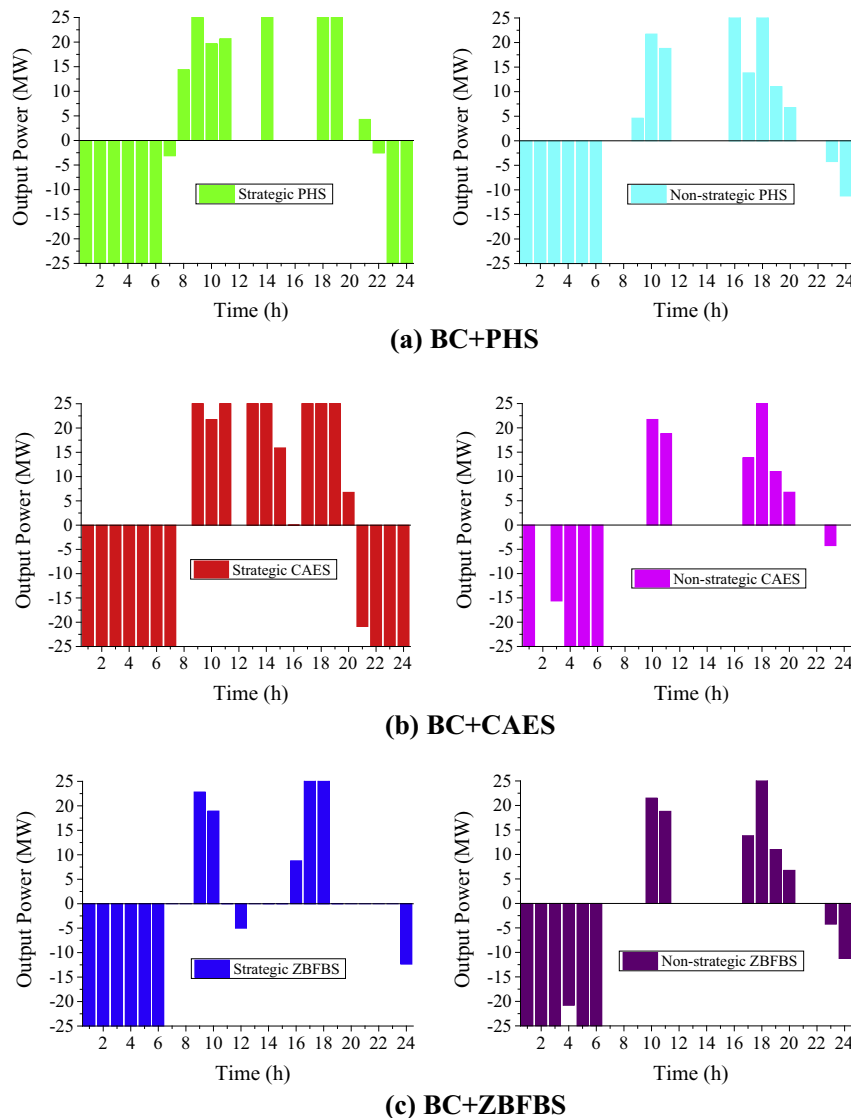


Fig. 4. The output power of PHS, CAES and ZBFBS at equilibrium.

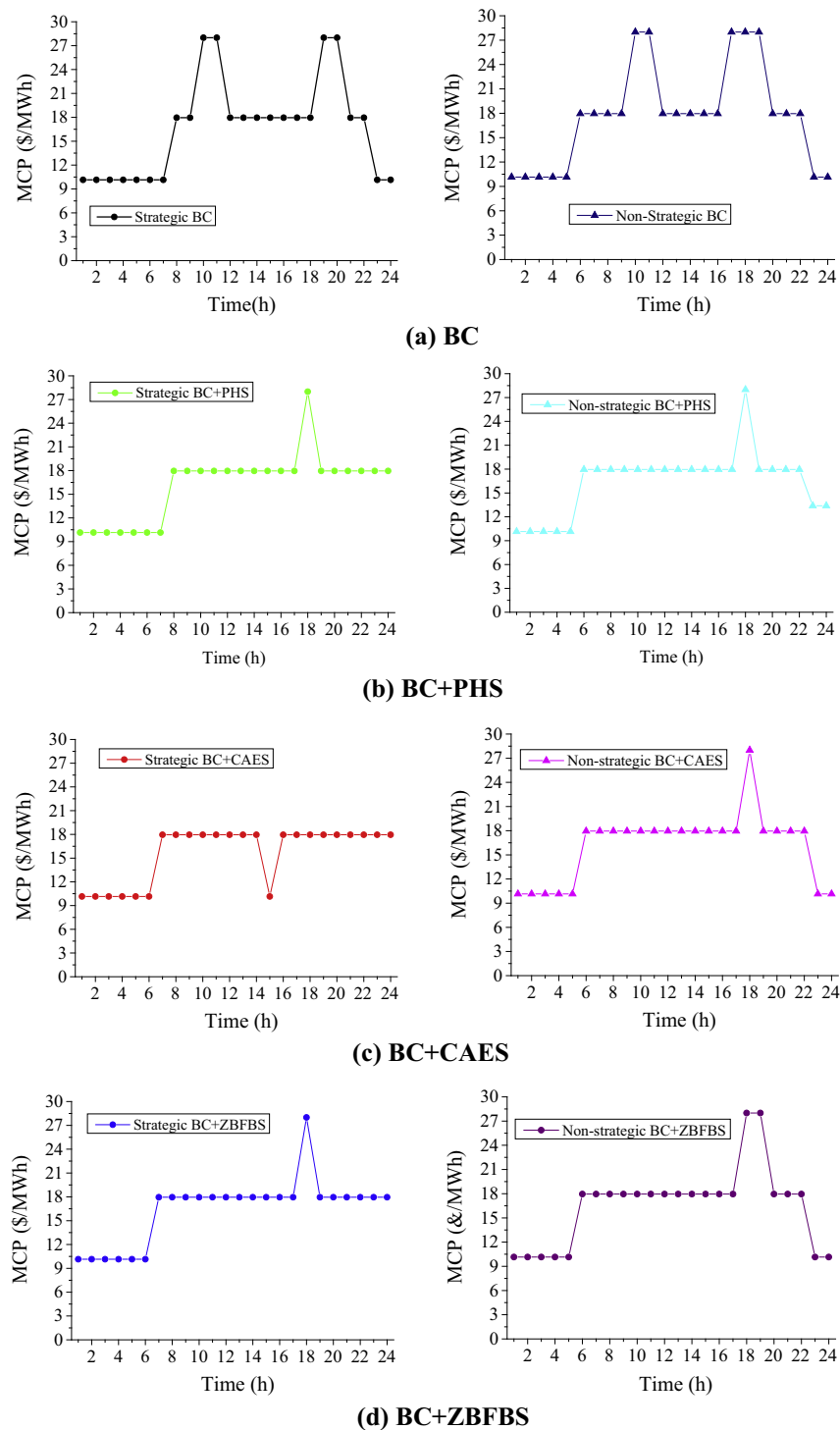


Fig. 5. The MCP curves of different cases.

Table 3

Comparisons of the average MCP, total consumer payment and ESS arbitrage profit in both strategic and competitive cases.

| Cases         |                             | BC                 | BC + PHS           | BC + CAES          | BC + ZBFBS         |
|---------------|-----------------------------|--------------------|--------------------|--------------------|--------------------|
| Strategic     | Average MCP (\$/MW h)       | 16.70              | 16.09              | 15.68              | 16.42              |
|               | Total consumer payment (\$) | $2.28 \times 10^5$ | $2.18 \times 10^5$ | $2.10 \times 10^5$ | $2.22 \times 10^5$ |
|               | ESS profit (\$)             | –                  | 609.70             | 122.13             | 221.56             |
| Non-strategic | Average MCP (\$/MW h)       | 17.77              | 16.36              | 16.09              | 16.51              |
|               | Total consumer payment (\$) | $2.43 \times 10^5$ | $2.21 \times 10^5$ | $2.18 \times 10^5$ | $2.24 \times 10^5$ |
|               | ESS profit (\$)             | –                  | 542.00             | 40.01              | 208.46             |

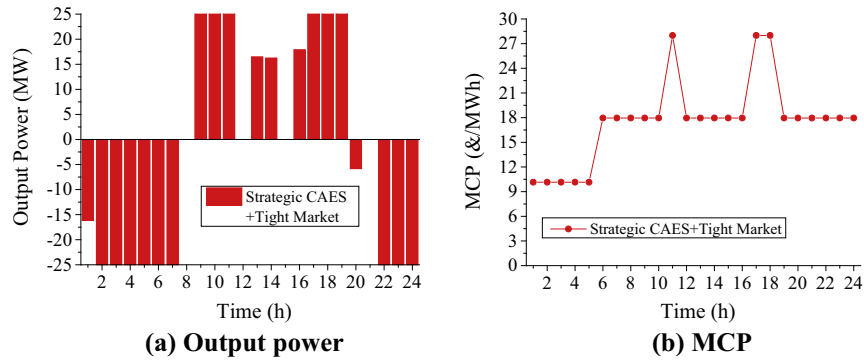


Fig. 6. Strategic behaviors of CAES change with the higher load.

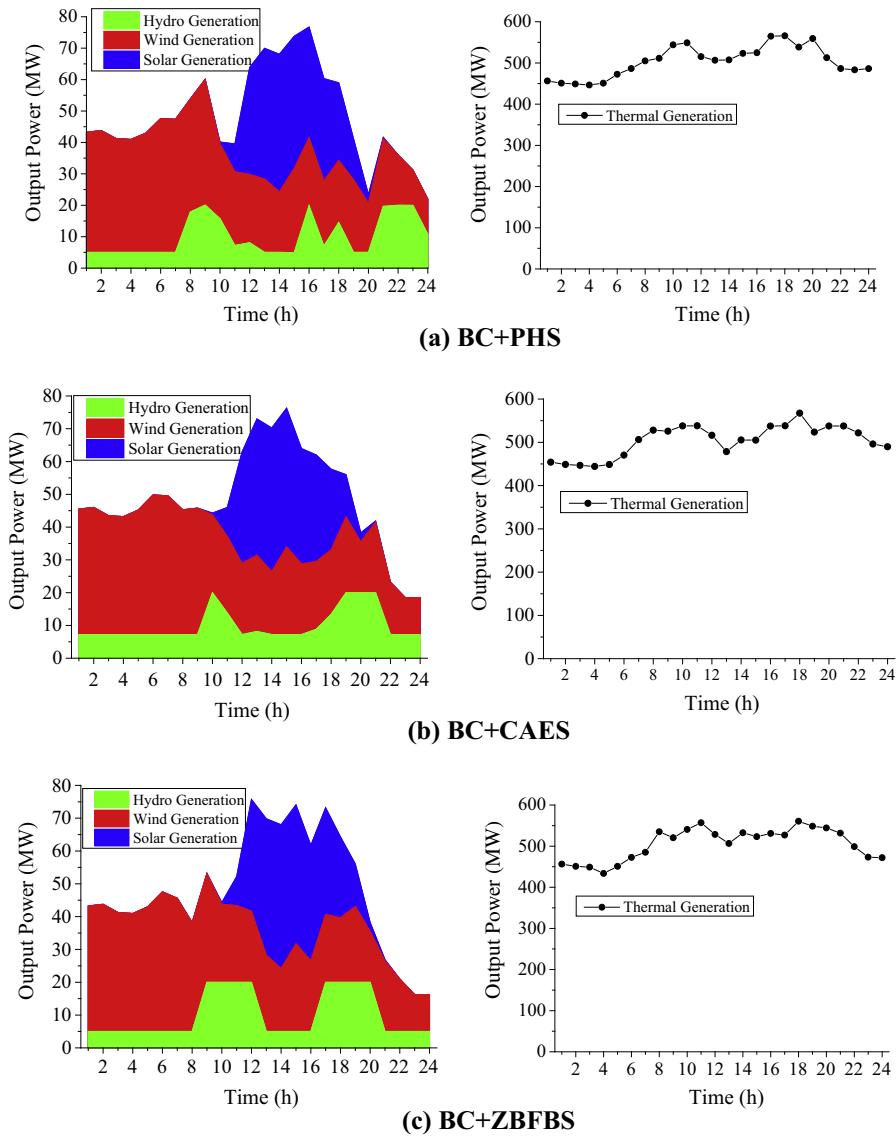


Fig. 7. Strategic interactions between ESSs and other generators.

which has the lowest energy limit and efficiency, charges and generates less than PHS and CAES. During the 12th hour the ZBFBS charges even with the comparatively high price in order to arbitrage in the following periods. As expected, the ESS that can

accommodate more energy and have higher efficiency has a stronger arbitrage capability in the market.

The non-strategic case in which all participants submit their true marginal costs are also included for comparison in Fig. 4. In

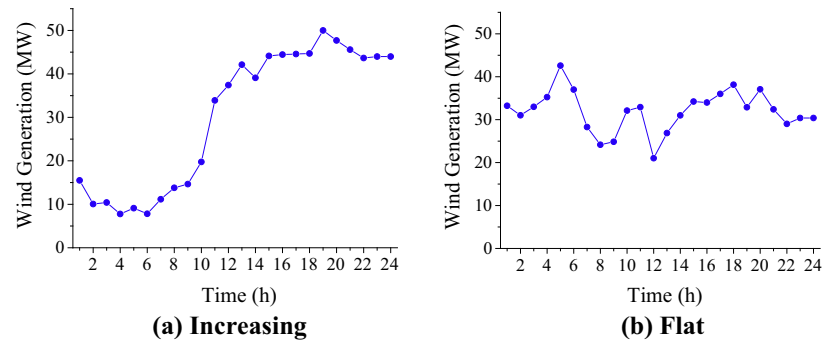


Fig. 8. Two wind generation profiles.

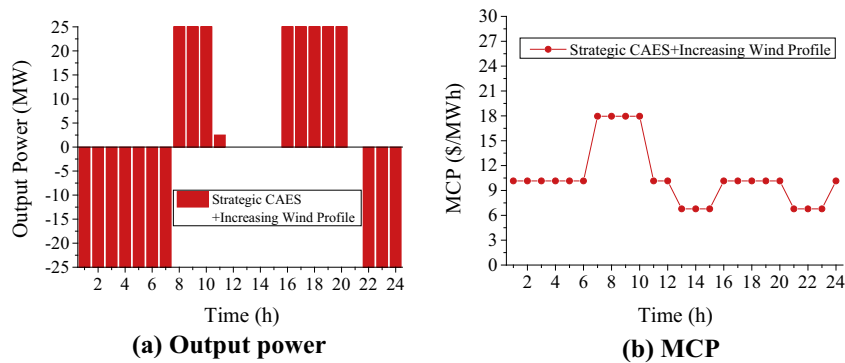


Fig. 9. Strategic behaviors of CAES change with the increasing wind profile.

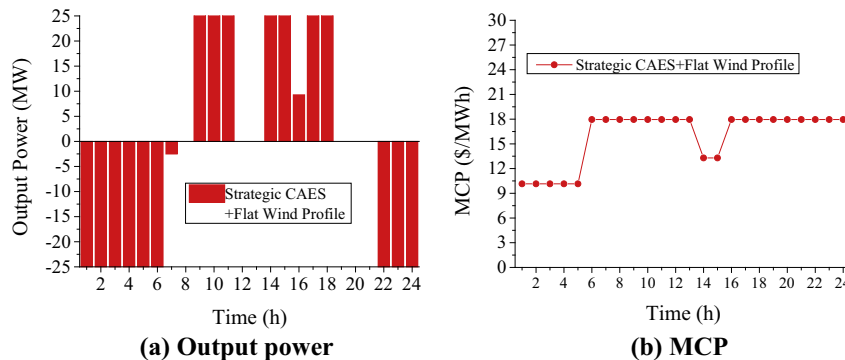


Fig. 10. Strategic behaviors of CAES change with the flat wind profile.

the non-strategic case, the ESSs are mainly dispatched by the system operator to achieve minimum total generation cost and thus the CAES is utilized less than PHS and ZBFBS because of its higher variable O&M cost. While in the strategic case all the storages speculatively choose to charge and discharge more than in the non-strategic case so that they can arbitrage as much as possible. The charged and discharged energies of ESSs in both strategic and non-strategic cases are presented in Table 2.

#### 6.4. Impacts of ESSs on the market equilibrium

Generally, the arbitrage behaviors of ESSs can achieve load shifting effects and accordingly peak shaving and levelling of the price curve in both the strategic case and the non-strategic case. However, the strategic behaviors of ESSs further level the price curve and result in average market prices lower than those of the

Table 4

Comparisons of the total payment of demand and CAES profit with different wind generation profiles.

| Strategic cases | Average MCP (\$/MW h) | Total consumer payment (\$) | CAES profit (\$) |
|-----------------|-----------------------|-----------------------------|------------------|
| CAES + D        | 15.68                 | $2.10 \times 10^5$          | 122.13           |
| CAES + I        | 10.61                 | $1.41 \times 10^5$          | 46.37            |
| CAES + F        | 15.93                 | $2.13 \times 10^5$          | 131.76           |

non-strategic cases, as shown in Fig. 5. For strategic BC without ESS, the average price is still lower than that of the competitive BC mainly because of the speculative behaviors of the cost-effective and high flexible hydro unit. For strategic CAES, the outcome exhibits longer production from 9th hour to 20th hour, decreasing the MCP further as compared to cases involving PHS and ZBFBS.

The impacts of the various ESSs on the total payment of the demand and the individual profits in both the strategic and non-strategic cases are presented in Table 3. In general, the higher the overall efficiency and the higher the energy the ESS can accommodate, the more strategic behaviors of ESSs and the lower the MCP and the total consumer payment.

PHS makes more profit than CAES and ZBFBS because of its lower variable O&M cost. While CAES obtains the lowest profit due to its high O&M cost, it actually has a higher capability to make profits than the PHS and ZBFBS, which can be seen from the profit differences between the strategic case and the competitive case. ZBFBS achieves a medium profit considering its moderate O&M cost although it has the lowest profit making capability. Therefore, each ESS maximizes its individual profit considering its energy limit, overall efficiency and its variable O&M cost. And the lower the variable O&M cost is, the more profit the ESS earns.

Considering CAES as an example, if the peak load goes up to 700 MW, the average MCP is \$17.58, i.e., a 12.1% increase, as shown in Fig. 6. The speculative behaviors of CAES result in higher profit due to the larger price difference between peak and valley.

Moreover, the strategic interactions between different kinds of ESSs and thermal units, hydro unit, wind farm and solar station are shown in Fig. 7. The renewables are most competitive and can be fully integrated considering the relatively sufficient flexibilities of the power system. The hydro unit is highly flexible to change its generations in different periods and can strategically interact with PHS, CAES and ZBFBS, collectively impacting the MCP and market equilibrium. The thermal units with high variable costs are less competitive and always obtain the last priority to generate. In other words, different kinds of ESSs mainly impact the strategic behaviors of hydro and thermal units.

### 6.5. Strategic interactions between ESS and wind units

Considering that the renewable generation has large uncertainties and variabilities, how the strategic behaviors of ESSs change with changes in the wind production profiles is considered below. Two typical wind generation profiles, that is, increasing profile and flat profile are illustrated in Fig. 8.

We consider CAES among the above three ESSs as an example. When the wind generation profile changes from decreasing (CAES + D), which is shown in Fig. 2, to increasing (CAES + I) or the flat pattern (CAES + F), the strategic behaviors of CAES and the MCP curve change accordingly, as presented in Figs. 9 and 10.

The comparisons of the total consumer payments and the CAES profits with different wind generation profiles are presented in Table 4. If the wind units generate as the increasing profile, the average MCP is lower than those of the other two profiles because the peak load can be partially provided by cost-effective wind power, which also shrinks the CAES profit due to the reduced peak-price periods. For CAES + F, the average MCP, the total consumer payment and CAES profit are the highest because the larger number of peak-price periods and the lower number of valley-price periods.

## 7. Conclusions

This paper formulates a multi-period EPEC model to investigate market equilibria considering the strategic behaviors of three typical ESSs and the interactions among other generators, including thermal, hydro, wind and solar units. The impacts of PHS, CAES and ZBFBS on the MCP, total consumer payment and their own net arbitrage profits are respectively compared. The case studies indicate that arbitrage behaviors of the ESSs will indirectly achieve the effects of load shifting, price levelling and a decrease on total

consumer payment. When various ESSs jointly compete in the electricity market, the ESS with higher efficiency and larger energy capacity dominates. In addition, when wind units generate with different profiles, the strategic behaviors of ESSs and the MCP curves change accordingly. The ESSs can obtain the highest profits with the flat wind generation profile and the lowest profits with the increasing profile.

Future work will involve transmission limit and unit commitment effects. Hopefully, this paper can provide some new ideas in the research of electricity market equilibrium.

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